Status	Finished
Started	Friday, 30 October 2026, 2:54 PM
Completed	Friday, 30 October 2026, 2:58 PM
Duration	4 mins 3 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting decreases the efficiency of selection operations
- b. Sorting is rarely used in programming
- c. Sorting makes it harder to search for items
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **2**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Dividing the list into halves
- c. Sorting the entire list in one pass
- d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **3**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Deleting data from a list
- c. Arranging data in a particular format
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Selection
c. Searching
d. Duplicates

The correct answer is: All of the above

Question **6**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 80,150,200,470,40,90,300,400,70,120
c. 40,70,80,90,120,150,200,300,400,470
d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It makes the analysis slower
- c. It has no effect on the analysis process
- d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **8**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It complicates the selection of items
- b. It makes it easier to select items based on their relationship to the rest of the items
- c. It makes the data unsorted
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **9**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **10**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By breaking the input into smaller parts and merging them
c. By comparing elements sequentially
d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **11**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching
b. Complexity
c. Merging
d. Sorting

The correct answer is:
Complexity

Question **12**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Dynamic programming
c. Greedy method
d. Divide and conquer

The correct answer is:
Divide and conquer

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Merge Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Linear Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It allows for faster searching
c. It has no effect on the search operation
d. It makes the search operation slower

The correct answer is: It allows for faster searching